TEXORA · COURSE NOTES · LECTURE SHEET

Sequences and Series

Infinite lists and infinite sums — when they settle to a limit, when they run away, and the tests that decide, ending with power series.

MATH-1101 · MATHEMATICS-I · TOPIC 2 · LEVEL 1 TERM 1

1. Sequences

A **sequence** is an ordered list of numbers $a_1, a_2, a_3, \ldots$, usually given by a formula for its n-th term a_n. The sequence **converges** to a limit L when its terms approach L as closely as desired for all sufficiently large n; otherwise it **diverges** (to $\pm\infty$, or by oscillation). Standard limits worth knowing: $1/n \to 0$; $r^n \to 0$ when $|r| < 1$ and diverges when $|r| \geq 1$ ($r \neq 1$); $(1 + 1/n)^n \to e$; $n^{1/n} \to 1$. A sequence that is both monotonic (always increasing or always decreasing) and bounded must converge — the principle behind many existence arguments.

2. Series and partial sums

A **series** is the sum of a sequence's terms, $a_1 + a_2 + \cdots$, written Σa_n. Since infinitely many additions cannot literally be performed, the sum is defined through the **partial sums** $s_N = a_1 + \cdots + a_N$: the series converges to S when the sequence of partial sums converges to S. One immediate and necessary test follows: **if Σa_n converges, then $a_n \to 0$.** The converse is false — the harmonic series $\Sigma\ 1/n$ has terms tending to 0 yet diverges, the standard warning example of this whole topic.

2.1 The geometric series

The most important explicit series sums a constant ratio: $a + ar + ar^2 + \cdots$. Its partial sums give the clean result

$$a + ar + ar^2 + \cdots = a / (1 - r), \quad \text{valid exactly when } |r| < 1.$$

WORKED EXAMPLE — A GEOMETRIC SUM

$3 + 1 + 1/3 + 1/9 + \cdots$ has $a = 3$ and $r = 1/3$, so the sum is $3 / (1 - 1/3) = 3 / (2/3) =$ **4.5**. The partial sums 3, 4, 4.33, 4.44, … visibly close in on 4.5.

2.2 The p-series

$\Sigma\ 1/n^p$ converges exactly when $p > 1$ and diverges when $p \le 1$. Thus $\Sigma\ 1/n^2$ converges, $\Sigma\ 1/n$ diverges ($p = 1$), and $\Sigma\ 1/\sqrt{n}$ diverges ($p = 1/2$). The p-series is the yardstick for the comparison tests below.

3. Tests for convergence (positive terms)

TEST	STATEMENT
n-th term test	If a_n does not tend to 0, the series diverges. (Inconclusive when $a_n \to 0$.)
Comparison test	If $0 \le a_n \le b_n$ and Σb_n converges, then Σa_n converges; if $a_n \ge b_n \ge 0$ and Σb_n diverges, then Σa_n diverges.
Ratio test	Let $L = \lim \lvert a_{n+1}/a_n \rvert$. $L < 1$: converges absolutely. $L > 1$: diverges. $L = 1$: inconclusive. Ideal for factorials and powers.
Root test	Let $L = \lim \lvert a_n \rvert^{1/n}$. Same conclusions as the ratio test; ideal when a_n is an n-th power.
Integral test	For a positive decreasing function f with $a_n = f(n)$: Σa_n and $\int f(x)\,dx$ converge or diverge together.

WORKED EXAMPLE — THE RATIO TEST

Test $\Sigma\ n! / 10^n$. The ratio of successive terms is $[(n+1)! / 10^{n+1}] \div [n! / 10^n] = (n + 1)/10$, which tends to ∞. Since $L > 1$, the series **diverges**: factorials eventually outgrow any fixed power.

4. Alternating series; absolute and conditional convergence

An **alternating series** has terms that alternate in sign, like $1 - 1/2 + 1/3 - 1/4 + \cdots$. By **Leibniz's test**, it converges when the absolute values of the terms decrease steadily to 0. A series is **absolutely convergent** when $\Sigma\lvert a_n \rvert$ converges (absolute convergence guarantees convergence), and **conditionally convergent** when it converges but $\Sigma\lvert a_n \rvert$ diverges. The alternating harmonic series is the standard example of conditional convergence: it converges (to $\ln 2$), while its absolute version, the harmonic series, diverges. Conditionally convergent series are delicate — rearranging their terms can change their sum — while absolutely convergent series behave like finite sums.

5. Power series

A **power series** $\Sigma\ c_n(x - a)^n$ is a series in which the terms involve powers of a variable. For each x it is an ordinary series, and the set of x for which it converges is an interval — the **interval of convergence** — centred at a with **radius of convergence** R, found from the ratio test applied to the terms: $R = \lim$

$|c_n / c_{n+1}|$ when the limit exists. Inside the interval the series defines a function that can be differentiated and integrated term by term. Familiar examples:

FUNCTION	POWER SERIES	VALID FOR
e^x	$1 + x + x^2/2! + x^3/3! + \cdots$	all x
sin x	$x - x^3/3! + x^5/5! - \cdots$	all x
cos x	$1 - x^2/2! + x^4/4! - \cdots$	all x
$1/(1 - x)$	$1 + x + x^2 + x^3 + \cdots$	$\|x\| < 1$
$\ln(1 + x)$	$x - x^2/2 + x^3/3 - \cdots$	$-1 < x \leq 1$

These **Taylor and Maclaurin series** (Maclaurin = Taylor about a = 0) are how calculators evaluate functions and how engineers replace awkward functions by polynomials near a working point.

6. Choosing a test — a working strategy

Faced with an unknown series, experienced solvers read its anatomy before reaching for a test. If the terms do not visibly tend to 0, stop: the series diverges. If the terms form a geometric pattern or resemble $1/n^p$, the two complete answers of §2 settle it directly. If factorials or n-th powers appear, the ratio test is built for them; if the whole term is an n-th power, the root test is quicker. If the terms are a rational or algebraic expression in n, compare against a p-series by keeping only the dominant powers (the limit-comparison form of the comparison test). If signs alternate, first test $\Sigma|a_n|$: absolute convergence ends the question, and only if that fails is Leibniz's test needed for conditional convergence. The worked habit matters more than any single test — state what you are testing, apply it cleanly, and conclude in words.

7. Key points and summary

- A series converges when its partial sums converge; its terms must then tend to 0, but terms tending to 0 do not guarantee convergence (harmonic series).
- Two series have complete answers: the geometric series (converges to $a/(1 - r)$ for $|r| < 1$) and the p-series (converges for $p > 1$).
- For other positive series use comparison, ratio, root or integral tests; for alternating series use Leibniz's test, and always check whether convergence is absolute or only conditional.
- Power series converge on an interval with a radius given by the ratio test, and inside it they can be differentiated and integrated term by term — the basis of Taylor and Maclaurin expansions.

8. Practice

Work each problem fully; final answers are given at the end.

1. Find $\lim (3n^2 + 1) / (n^2 + 5)$ as $n \to \infty$.
2. Sum the geometric series $5 + 5/2 + 5/4 + 5/8 + \cdots$.
3. Decide, with the test named: does $\Sigma\ 1/\sqrt{n}$ converge?
4. Apply the ratio test to $\Sigma\ 2^n/n!$ and state the conclusion.
5. Classify $\Sigma\ (-1)^{n+1}/n$ as absolutely convergent, conditionally convergent, or divergent, with reasons.
6. Find the radius of convergence of $\Sigma\ x^n/n$.

Answers

1. 3 (divide top and bottom by n^2).
2. $a = 5$, $r = 1/2$, sum $= 5 / (1 - 1/2) = 10$.
3. Diverges — it is a p-series with $p = 1/2 \le 1$.
4. Ratio $= 2/(n + 1) \to 0 < 1$, so the series converges (absolutely).
5. Conditionally convergent: Leibniz's test gives convergence, but $\Sigma\ 1/n$ (the absolute series) is the divergent harmonic series.
6. $R = 1$ (the ratio of successive coefficients tends to 1).